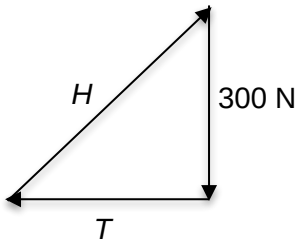
Tampines Meridian Junior College

2026 JC2 H2 Physics Preliminary Examination Paper 2

Suggested Solution

1 (a)



[M1] for correct vector triangle with correct direction
[A1] for correct labels of forces
[deduct one mark for label angle 30°]

(b)

By taking moments about the hinge,
$T \times 0.60 = 300 \times 1.5 \cos 30^\circ$ [C1]
$T = 650$ N [A1]

(c)

Horizontally, $\Sigma F_x = 0$
$T = H_x$,
$H_x = 650$ N

Vertically, $\Sigma F_y = 0$
$H_y = 300$ N [C1 for both equations]

$H = \sqrt{H_x^2 + H_y^2} = \sqrt{650^2 + 300^2}$
$= 716$ N [A1]

$$\theta = \tan^{-1}\left(\frac{H_y}{H_x}\right)$$
$$= \tan^{-1}\left(\frac{300}{650}\right)$$
$$= 24.8^\circ$$ [A1]

24.8° anticlockwise above the horizontal.

2	**(a)**	The area under F_E represents the work done by the driving force of the car's engine. [A1]	
	(b)	**(i)**	Net work done = area bounded by F_E and F_R $= \frac{1}{2}(50)(4000) = 1.0 \times 10^5$ J [C1] This is also the gain in the kinetic energy of the car. Since the car started from rest, the maximum speed of the car is: $\frac{1}{2}(1800)v^2 = 100000$ [M1] $v = 10.5$ m s^{-1} [A1]
		(ii)	After d = 50 m, the car is at maximum speed. The power of the car at that point is $P = Fv = (1000)(10.54)$ [M1] $= 1100$ W [A1] Ecf allowed
	(c)		Since the maximum speed is unchanged but the mass has increased, the final kinetic energy of the car is increased. Since the power of the engine is the same, more time is now needed to for the car to gain this larger amount of kinetic energy.

3	**(a)**	Vertical component of the initial velocity, $u_y = 0$ $s_y = \frac{1}{2}a_y t^2$ $= \frac{1}{2}(9.81)(2.5)^2$ $= 31 \text{ m}$	[M1] [M1] [A0]
	(b)	$u_x = v_x = \frac{240}{2.5} = 96.0 \text{ m s}^{-1}$ $v_y = a_y t = (9.81)(2.5) = 24.5 \text{ m s}^{-1}$ Resultant velocity, $v = \sqrt{(96.0)^2 + (24.5)^2} = 99.1 \text{ m s}^{-1}$ Direction, $\theta = \tan^{-1}\frac{24.5}{96.0} = 14.3^\circ$ clockwise below horizontal	[C1] [C1] [A1] [A1]
	(c)	Even though the initial speed had increased, the initial vertical component of the speed remains the same. Hence there is no change to the time of flight.	[M1] [A1]

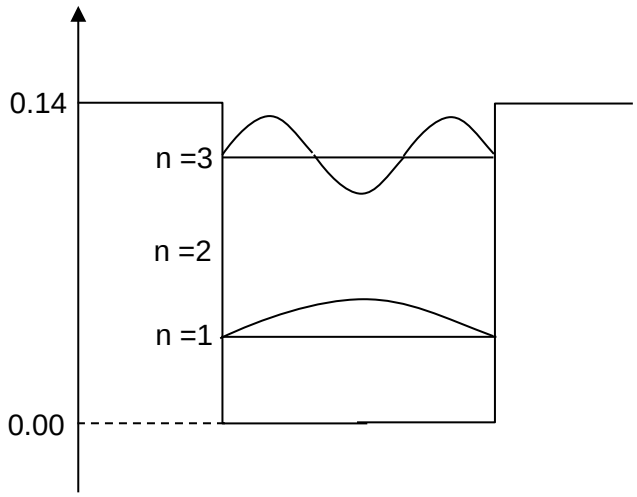
4	**(a)**	Simple harmonic motion (SHM) is defined as the oscillation of a particle such that its <u>acceleration is proportional to its displacement</u> from the equilibrium position, and is always <u>directed towards the equilibrium position</u> (or opposite in direction to the displacement).		B1 B1
	(b)	**(i)**	$\omega = 2\pi f = 2\pi(3.0) = 18.85 \text{rads}^{-1}$ Maximum kinetic energy = ½ $m\,\omega^2\,x_o^2$ = ½ (0.80)(18.85) ^{2}(0.016)^2 = 0.0364 J ≈ 0.036 J	C1 A1
		(ii)	$x = -x_0 \cos \omega t$ $v = \frac{dx}{dt} = \omega x_0 \sin \omega t$ $\omega = 2\pi f = 2\pi(3.0) = 18.85 \text{rads}^{-1}$ Velocity v = (18.85)(0.016) sin(18.85(0.20)) Velocity v = -0.1773 m s^{-1} Speed v ≈ 0.18 m s^{-1} (Do not accept negative sign for final answer.)	C1 A1

5	**(a)**	Increase in internal energy = work done on the system + heat supplied to the system. One mark for correct definition [M1] One mark for putting into a "word equation" [A1]			
	(b)	**(i)**	$u_C = u_A + \Delta u$ $= 45 + (-40 + 90)$ $= 95$ kJ		[C1] [A1]
		(ii)	$\Delta u = Q + W$ $-40 + 90 = Q - 20$ $Q = 70$ kJ		[C1] [A1]
		(iii)	**1.**	net work done on gas = -40 + 31 = -9.0 kJ	[C1] [A1]
		(iii)	**2**	Since there is no change in internal energy in a cycle, $\Delta u = 0$ $Q_{net} + W_{net} = 0$ $Q_{net} - 9.0 = 0$ $Q_{net} = 9.0$ kJ Ecf allowed.	[A1] [A1]

6	**(a)**	Magnetic flux is defined as the product of magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density. (from LO) Or Magnetic flux is the product of the flux density normal to the surface and the area of the surface. (from notes). [1]	
	(b)	**(i)**	90° or 270° [1]
		(ii)	Max Φ = NBA $= 300 \times 5.0 \times 10^{-2} \times \pi(4.0 \times 10^{-2})^2$ [C1] = 0.0754 Wb [A1]
		(iii)	Average emf $= \frac{\Delta \Phi}{\Delta t}$ $= \frac{(0.0754-0)}{0.50 \times 10^{-3}}$ [C1] $= 151\,\text{V}$ [A1]
		(iv)	The minimum e.m.f. can be determined from the time at which the gradient of the graph is zero. [1] The first instant is at t = 0.50 ms. (no marks deducted for sf) [1]
		(v)	As the coil rotates, the magnetic flux linkage of the coil varies sinusoidally. By Faraday's Law, the induced e.m.f is directly proportional to the rate of change of magnetic flux linkage. [M1] Hence the e.m.f will also vary sinusoidally. [A1]

7	**(a)**	The horizontal intercept of the graph shows the <u>threshold wavelength</u> above which no photoemission of electrons will take place. [A1] Accept <u>threshold frequency</u> if a reference is made to $v = f\lambda$ Also accept: <u>Maximum wavelength</u> above which <u>no emission</u> of electrons will take place.	
	(b)	**(i)**	The minimum energy required can be found when the photon with the threshold wavelength causes photoemission – this is the x-axis intercept of the graph. Photon energy at threshold wavelength $= \frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{4.2 \times 10^{-7}}$ [C1] $= 4.74 \times 10^{-19}$ J $= 4.7 \times 10^{-19}$ J [A1]
		(ii)	The kinetic energy of the emitted electron will be: $KE = \frac{hc}{\lambda} - \phi$ $= \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{2.0 \times 10^{-7}} - 4.74 \times 10^{-19}$ [C1] $= 5.21 \times 10^{-19}$ J [A1]
	(c)	Light is made of discrete <u>quanta of electromagnetic radiation</u> / <u>packets of energy</u> called <u>photons.</u> [B1] <u>One electron</u> can only <u>absorb</u> the energy of <u>one photon</u> at a time. [B1] Photons with wavelengths <u>longer than the threshold wavelength</u> / <u>frequency lower than the threshold frequency</u> <u>do not have sufficient energy</u> [B1] for electrons to overcome the surface energy barrier to escape the metal.	

8	**(a)**	Number of silicon atoms per m^3: $n_{silicon} = \frac{2.33 \times 10^3}{0.0281} \times 6.02 \times 10^{23} = 4.99 \times 10^{28}$ [C1] Number of phosphorus atoms per m^3: $n_{phosphorus} = \frac{4.99 \times 10^{28}}{2 \times 10^6}$ [M1] $= 2.50 \times 10^{22}$ [A0]
	(b)	Using $I = nAvq$ $0.012 = 2.5 \times 10^{22} \times 0.25 \times 10^{-2} \times 1.1 \times 10^{-2} \times v \times 1.6 \times 10^{-19}$ [M1] $v = 0.109 \text{ m s}^{-1}$ [A1]
	(c)	The electric field strength is the negative of the potential gradient. Since the potential is <u>decreasing towards the right</u>, or <u>Fig. 8.3 showed negative gradient (together with electric field strength is the negative of the potential gradient)</u>, [M1] the electric field is pointing <u>left to right</u>. [A1]
	(d)	$E = \left\lvert \frac{\Delta V}{\Delta x} \right\rvert$ $= \frac{0.44 - (-0.34)}{(0.59 - 0.26) \times 10^{-6}}$ [M1] $= 2.36 \times 10^6 \text{ V m}^{-1}$ [A1]
	(e)	$C = \frac{Q}{V}$ $= \frac{3.6 \times 10^{-8}}{0.44 - (-0.34)}$ [M1] $= 4.62 \times 10^{-8} \text{ F}$ [A1]
	(f)	minimum wavelength: $\lambda = \frac{hc}{(0.29 + 1.42 + 0.34)e}$ [M1] $= 606 \times 10^{-9} \text{ m}$ [A1]
	(g)	<u>red/orange light</u> for 606 nm [B1] <u>infrared</u> for 875 nm [allow ecf from part (e)]

	(h)	**(i)**	energy level / eV  [B1] for n = 1 [B1] for n = 3
		(ii)	kinetic energy $E = \frac{p^2}{2m}$ [C1] $E = \frac{\left(\frac{nh}{2d}\right)^2}{2m}$ [M1] $E = \frac{n^2h^2}{8md^2}$ [A0]
		(iii)	When n = 3 $E = \frac{(3)^2\left(6.67\times10^{-34}\right)^2}{8\times9.11\times10^{-31}\times\left(5\times10^{-9}\right)^2}$ [M1] $=2.17\times10^{-20}$ $=0.136$ eV [A1] This is very close to the potential height of the quantum well of 0.14 eV. The <u>next energy level</u> is greater, almost double due to the coefficient of 4^2 and so <u>will exceed the energy barrier</u>, and the electron with that amount of energy will not be confined in the well. [B1]